# Swarm Witness: What Do Agent-Safety Mechanisms Actually Establish?

**Tracks:** 3 (Strengthening Agent Infrastructure), 4 (Multi-Agent Oversight and Control)

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### Abstract

Multi-agent safety mechanisms are often described as novel when they implement durable authorization, crash-consistent recovery, or evidence packaging. We ask: after matching policy, observations, trust assumptions, durability, and enforcement coverage, which claimed benefits remain? We contribute Swarm Witness, an executable claim-audit method that pins third-party artifacts, states precise claim cards, runs discriminating checks against classical baselines, and records fail-closed conclusion labels. On three pinned targets we obtain: SoundGate recovery-barrier (R1) reproduced — 153 histories/policy, original prohibited effects 143 vs barrier 0, with 147 strict refusal-startup refusals; EvidenceGraph receipt stress (R2) reproduced — 36/36 condition-by-seed checks, old-rule relay wrong attributions [1,1,1] per seed vs safer supported [0,0,0], and clean withheld-without-exclusivity [12,12,12]; BMO retained-grant vs ClassicalGate (R3) assumption_dependent — 6/6 preferred Go tests pass, 3/3 proxy rows ok, 0 isomorphic decision matches across 4 comparison rows. The takeaway is that two widely circulated "agent-safety" repairs restore classical recover-before-admit and no-implicit-exclusivity controls under published harnesses, while a third durable-grant story shows mechanism-class affinity only through proxies—not isomorphic decision equivalence. Auditors should publish matched baselines and labels, not treat branding as establishment.

## 1. Introduction

Agent-facing "safety" components—authorization gates, receipt graphs, retained grants—are frequently marketed as multi-agent innovations. Many instantiate older durable-log, provenance, and admit-barrier patterns. When a report claims a restart-safe refusal, receipt-based attribution, or retained-grant relaunch control, auditors need to know whether the result is a new property, a repaired instance of a classical mechanism, or an assumption-dependent analogy.

**Research question.** After matching policy, observations, trust assumptions, durability, and enforcement coverage, which claimed benefits of agent-safety mechanisms remain?

**Threat / failure modes.** (i) An admission gate forgets an acknowledged rejection across restart when the durable log is not recovered before admit. (ii) A verifier treats possession of a copied receipt token as event attribution without exclusivity or authenticity. (iii) Retained start/grant metadata is treated as standing relaunch authority without consulting a durable grant table.

**Main contributions.**

1. **Swarm Witness claim-audit method:** pinned artifacts, typed claim cards, classical baselines, discriminating experiments, and fail-closed labels (`reproduced`, `assumption_dependent`, …) with an evidence index checker.

2. **Executable audits of three public/pinned targets** (SoundGate recovery barrier; EvidenceGraph receipt stress; BMO retained-grant owner path) with metrics taken only from committed `evidence/*/summary.json`.
3. **Classical baseline cards** showing when a "novel" repair is recover-before-admit or no-implicit-exclusivity reasserted, versus when mechanism-class proxies do not justify isomorphic decision claims.

Tracks: **3** (Strengthening Agent Infrastructure) and **4** (Multi-Agent Oversight and Control). Preexisting BMO code and Oct 25 scaffold are marked in `PREEXISTING.md`; judged weekend work is the audit method, pins, baselines, experiments, and evidence packaging.

## 2. Related Work

We build on Apart Swarm Alignment themes of identity, durable commitments, and oversight under copy/rename/relay (Hammond et al., 2025; Chan et al., 2025; Greenblatt et al., 2024). Concrete targets:

- **SoundGate / "A Restart Is Not Permission"** — durable auth decisions and recovery-barrier repair (pinned package `soundgate==0.1.3`, source revision `6568403`).
- **EvidenceGraph / Junejo "Receipt Possession Is Not Event Attribution"** — stress harness distinguishing possession from attribution (git rev `a5ea5c9`, paper baseline `efbd419`).
- **BMO ordinary-retry retained grant** — preexisting durable grant / owner-witness tests under pinned `bmo_root` (read-only `go test`).

Classical nulls (baselines in-repo): durable log validate/repair/replay before admit; no implicit exclusivity for receipt tokens; durable grant table checked before admit with retained metadata never authorizing relaunch. Swarm Witness differs from building a new gate: it asks what labels survive matched baselines and discriminating checks.

## 3. Methods

**Claim cards** (`claims/*.yaml`) fix `property`, `scope`, `quantifier`, `success_criterion`, `assumptions`, `baseline`, and `discriminating_checks`. **Pins** (`artifacts/pins.yaml`) freeze revisions and hashes. **Baselines** state the mechanism class without equating every proxy to a decision match. **Evidence** uses `ResultRecord` labels; `scripts/check_evidence.py` fails closed if `evidence/index.yaml` disagrees with `result.yaml` or files are missing.

**R1 (SoundGate).** Rebuild and sweep the paper recovery-barrier harness at pinned source `6568403`. Compare policies `original`, `strict`, and `barrier` over the published histories. Success for reproduction: barrier `prohibited_effects == 0` with original showing the defect. Supplemental cargo `wal_recovery_barrier` on 0.1.4 recorded in notes.

**R2 (EvidenceGraph).** Run `incident_stress.py` at `a5ea5c9` against baseline `efbd419`. Contract checks over seeds `{0,1,2}`: old-rule wrong attributions on relay, safer supported/correct on relay, clean withheld without exclusivity, false-exclusivity confident errors. Label `reproduced` when all contract checks pass and the safer rule restores abstention under undeclared exclusivity.

**R3 (BMO vs ClassicalGate).** Run preferred retained-grant Go tests read-only; compare four ClassicalGate conditions to BMO observations as `mechanism_class_proxy`, `proxy_mismatch`, or isomorphic match. Do **not** hardcode BMO decisions from ClassicalGate. Label `assumption_dependent` when affinity is proxy-only (0 isomorphic matches).

## 4. Results

Labels from `evidence/index.yaml` / `result.yaml` only: R1 **reproduced**, R2 **reproduced**, R3 **assumption_dependent**. Metrics below cite `evidence/*/summary.json`.

### 4.1 R1 — SoundGate recovery barrier (`reproduced`)

Fresh local rebuild+sweep; `all_checks_pass: true`; `raw_sha256=51b4acd6…`. Histories per policy: **153**.

| Policy | cases | prohibited_effect | refused_before_ack | supported | fresh_work |
|---|---|---|---|---|---|
| original | 153 | **143** | 4 | 6 | 149 |
| strict | 153 | 0 | **147** | 6 | 6 |
| barrier | 153 | **0** | 0 | **153** | 153 |

Barrier restores acknowledge-and-restrict across restart under the published histories; original loses restrictions (143 prohibited effects). Cargo `wal_recovery_barrier` on 0.1.4: 3 passed (notes). Mechanism class remains classical recover-before-admit (`baselines/r1_wal_recover_before_admit.md`).

### 4.2 R2 — Receipt ≠ attribution (`reproduced`)

`condition_by_seed_checks` **36**, passed **36**; seeds 0,1,2; `all_contract_checks_pass: true`; `source_tree_sha256=2962568f…`.

| Metric (per seed) | seed 0 | seed 1 | seed 2 |
|---|---|---|---|
| old_rule wrong attributions on relay | 1 | 1 | 1 |
| safer supported on relay | 0 | 0 | 0 |
| safer correct on relay | 0 | 0 | 0 |
| clean withheld without exclusivity | 12 | 12 | 12 |
| shared_tokens correct | 0 | 0 | 0 |
| false_exclusivity confident errors | 1 | 1 | 1 |

Relay seed-0 outcomes: ambiguous 10, unmatched 2. Safer default withholds attribution without exclusivity; old rule errs once per seed on relay. Mechanism class: no implicit exclusivity (`baselines/r2_no_implicit_exclusivity.md`).

### 4.3 R3 — Retained grant vs ClassicalGate (`assumption_dependent`)

BMO Go tests: **pass_count 6**, **fail_count 0**, `package_ok: true`. Comparison rows: **proxy_rows_ok 3 / 3**, **isomorphic_decision_matches 0 / 4**.

| Condition | ClassicalGate | BMO observed | Relation |
|---|---|---|---|
| retained_start_claim + relaunch | deny | not_isomorphic | mechanism_class_proxy |
| rejected + restart + execute | deny | not_isomorphic | proxy_mismatch |
| active_grant + execute | allow | not_isomorphic | mechanism_class_proxy |
| stale grant copy vs current | deny | not_isomorphic | mechanism_class_proxy |

Proxy rows show deny/allow-shaped affinity (spent/prior-consumer, owner-witness admit, wrong/altered grant). `rejected+restart` maps to process-reload in the null model; BMO

`LegacyWriterBoundary` is a writer-boundary race (`proxy_mismatch`)—not claimed as a decision match.

## 5. Discussion

**Implications.** For Tracks 3–4, durable identity and oversight claims should be scored against classical admit and evidence rules. R1 and R2 show that published repairs can **reproduce** restorations of those rules under pinned harnesses; that is valuable engineering and audit evidence, not proof of a new multi-agent primitive. R3 shows how easy it is to over-claim: related tests and proxy shapes ≠ isomorphic contracts.

**Limitations (summary).** Finite pinned sweeps ≠ general equivalence across deployments; R3 is explicitly assumption-dependent; preexisting BMO surfaces were consulted read-only. Full Limitations and Dual-Use appendix follows.

**Future work.** Expand pins to more gate implementations; strengthen R3 toward shared observation APIs where isomorphic matches are possible; automate baseline-vs-claim diff reports for continuous oversight of agent infrastructure.

## References


Chan, A., et al. (2025). *Infrastructure for AI Agents*.

Greenblatt, R., et al. (2024). *AI Control* protocols / multi-agent oversight context.

Hammond, L., et al. (2025). *Multi-Agent Risks from Advanced AI*.

Junejo / EvidenceGraph. *Receipt Possession Is Not Event Attribution* (Apart project; pinned git revs in `artifacts/pins.yaml`).

SoundGate. *A Restart Is Not Permission: When an Agent Gate Forgets No* (Apart project; `soundgate==0.1.3`, source `6568403`).

Apart Research. Swarm Alignment Hackathon Guidelines (2026). Template: https://docs.google.com/document/d/1_1_27EyZriMlc8n8EBE2YScaeyMjm-8xjR097nmHXAE/copy?usp=sharing


## Appendix A. Limitations and Dual-Use / Ethical Considerations

### What each result does and does not establish

**R1 (`reproduced`).** Establishes that, on the pinned SoundGate recovery-barrier harness (source revision `6568403`), the barrier policy yields `prohibited_effects = 0` across 153 histories while the original policy yields 143 prohibited effects, with 147 strict refusal-startup refusals under the strict policy. Does **not** establish that every agent admission gate is restart-safe, that SoundGate is free of other defect classes, or that recover-before-admit is a new multi-agent invention (it matches the classical durable-log baseline).

**R2 (`reproduced`).** Establishes that the pinned EvidenceGraph stress pack (`a5ea5c9` / baseline `efbd419`) passes 36/36 condition-by-seed checks: old-rule relay wrong attributions are 1 per seed; safer relay supported attributions are 0; clean runs withhold all 12 attributions without exclusivity declarations. Does **not** establish that all receipt or ledger systems are safe under relay/copy, or that exclusivity declarations themselves cannot be gamed outside the modeled threat surface.

**R3 (`assumption_dependent`).** Establishes that preferred BMO retained-grant tests pass (6/6) and that ClassicalGate vs BMO comparison yields 3/3 ok proxy rows with **0** isomorphic decision matches across 4 rows (one `proxy_mismatch`). Does **not** establish that ClassicalGate

and BMO share an API or that deny/allow-shaped proxies imply decision equivalence. Retained metadata ≠ relaunch authority remains a classical durable-grant-table claim, shown here only via mechanism-class affinity under stated assumptions.

## Finite experiments are not general equivalence

Pinned histories, seeds `{0,1,2}`, and preferred Go tests are finite. Passing them does not imply equivalence across versions, storage backends, concurrent writers outside the model, or production deployments. Labels apply to the stated scope on the claim cards only.

## Dual-use / adversarial use of the audit method

An executable claim-audit method can help **defenders** find weak gates and attribution gaps—and can also help **adversaries** locate the same gaps (restart races, possession-as-attribution, retained-metadata relaunch). We publish **tests, pins, matched baselines, and labels**, not exploit kits, weaponized harnesses, or production credentials. Reproduction artifacts are scoped to already-public or author-local preexisting trees (`PREEXISTING.md`).

## Preexisting code vs weekend analysis

Oct 25 scaffold paths and the external BMO tree at pinned `bmo_root` predate this sprint and were not modified for R3 beyond read-only `go test`. Weekend work is the Swarm Witness schema, pins, claim cards, baselines, experiment wrappers, evidence records, index checker, and this report. Judges should score the audit contribution, not treat preexisting BMO surfaces as sprint inventions.